Unit-2.

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Equivalence & Minimization of DFA

1. Identify distisguishable / indistinguishable states.

Def:

Distinguishable & Indistinguishable:

Two states p & q of a DFA are equivalent or indistinguishable if and only if $\delta(p,w)$ and $\delta(q,w)$ are final states or both $\delta(p,w)$ & $\delta(q,w)$ are ~~not~~ non-final states for all $w \in \Sigma^*$
i.e; i) if $\delta(p,w) \in F$ & $\delta(q,w) \in F$, then the states p and q is indistinguishable
ii) if $\delta(p,w) \notin F$ & $\delta(q,w) \notin F$ then also the states p and q are indistinguishable

1. Obtain the distinguishable table for the automaton shown below and Minimize the DFA.

states	0	1
→ A	B	F
B	G	C

automaton → one machine

automata → more machine

(C)	A	C
D	C	G
E	H	F
F	C	G
G	G	E
H	G	C

= ~~(A,B) (A,C) (A,D) (A,E) (A,F) (A,G)~~ (A,H)

(A,B) (A,C) (A,D) (A,E) (A,F) (A,G) (A,H)
(B,C) (B,D) (B,E) (B,F) (B,G) (B,H)
(C,D) (C,E) (C,F) (C,G) (C,H)
(D,E) (D,F) (D,G) (D,H)
(E,F) (E,G) (E,H)
(F,G) (F,H)
(G,H)

B	X						
C	X	X					
D	X	X	X				
E		X	X	X			
F	X	X	X		X		
G	X	X	X	X	X	X	
H	X		X	X	X	X	X
	A	B	C	D	E	F	G

X → step1

X → step2

X → step 2 ii)

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Step 1:

For each pair (p, q) where $p \in Q$ & $q \in Q$. if $p \in F$ & $q \notin F$ or viceversa, then the pair (p, q) is distinguishable & mark the pair by putting 'x'

Step 2:

For each pair (p, q) and for each $a \in \Sigma$ find $\delta(p, a) = r$ and $\delta(q, a) = s$. if the pair (r, s) is already marked as distinguishable, then the pair (p, q) is also distinguishable and mark it as say 'x'. 'x'
Repeat step 2 until no previously unmarked pairs are marked.

(p,q)	δ, 0	s 1
(A,B)	(B,G)	(F,C)
(A,D)	(B,C)	(F,G)
(A,E)	(B,H)	(F,F)
(A,F)	(B,C)	(F,G)
(A,G)	(B,G)	(F,E)
(A,H)	(B,G)	(F,C)
(B,D)	(G,C)	(C,G)
(B,E)	(G,H)	(C,F)
(B,F)	(G,C)	(C,G)
(B,G)	(G,G)	(C,E)

(B, H)	(G, G)	(C, C)
(D, E)	(C, H)	(G, F)
(D, F)	(C, C)	(G, G)
(D, G)	(C, G)	(G, E)
(D, H)	(C, G)	(G, C)
(E, F)	(H, C)	(F, G)
(E, G)	(H, G)	(F, E)
(E, H)	(H, G)	(F, C)
(F, G)	(C, G)	(G, E)
(F, H)	(C, G)	(G, C)
(G, H)	(G, G)	(E, C)

δ	0	1
(A, E)	(B, H)	(F, F)
(A, G)	(B, G)	(F, E)
(B, H)	(G, G)	(C, C)
(D, F)	(C, C)	(G, G)
(E, G)	(H, G)	(F, E)

B, G → is marked (E, G) can be marked ×

any one is marked after pervious table then S(p, q) can be marked

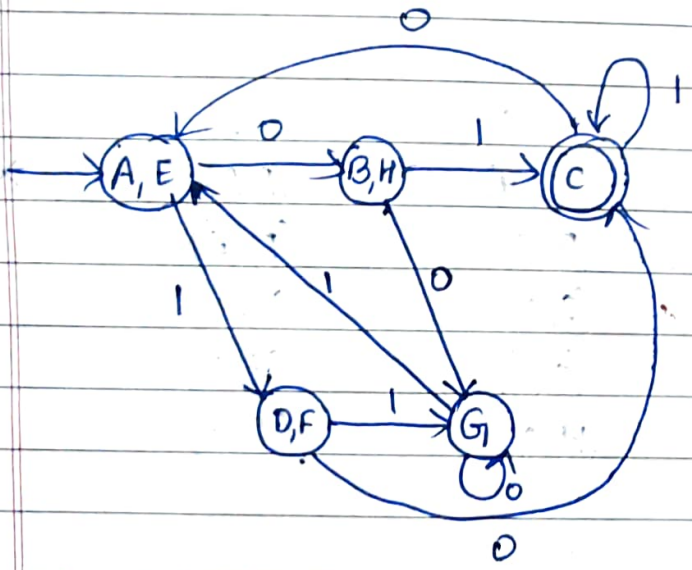
Observe that the pairs (A, E), (B, H) and (D, F) are not all marked and are considered to be indistinguishable pair states.

→ The unmarked pairs (A, E) (B, H), (D, F) are indistinguishable and the states C & G are distinguishable.

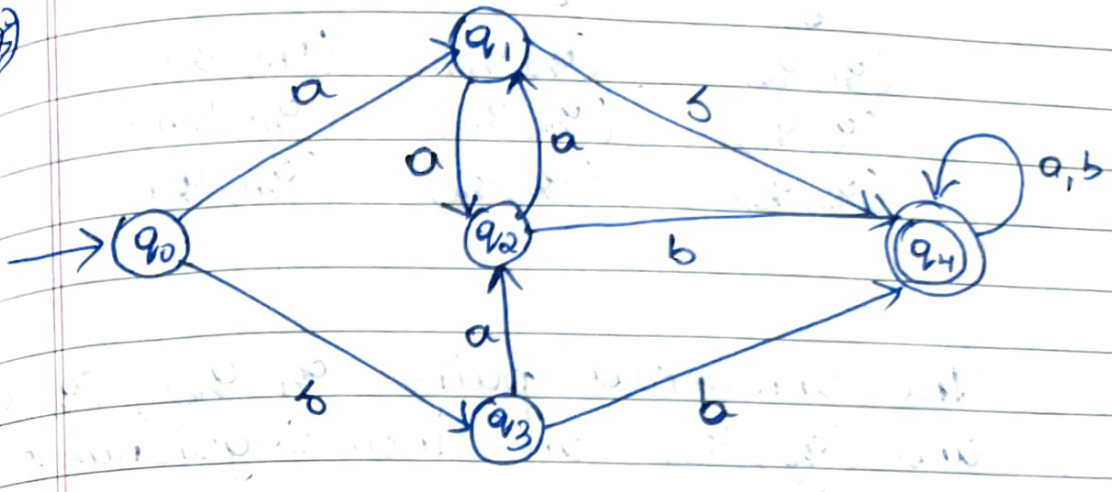
The various group that represents states of minimized DFA are:
(A,E), (B,H), C, (D,F), G

δ	0	1
→ (A,E)	(B,H)	(D,F)
(B,H)	G	C
* C	(A,E)	C
(D,F)	C	G
G	G	(A,E)

(F,F) is not there in minimized DFA so the equivalent state is (D,F)



2] Minimize the following DFA



$= (q_0, q_1)\ (q_0, q_2)\ (q_0, q_3)\ (q_0, q_H)$
$(q_1, q_2)\ (q_1, q_3)\ (q_1, q_H)$
$(q_2, q_3)\ (q_2, q_H)$
(q_3, q_H)

q_1	×			
q_2	×			
q_3	×			
q_H	×	×	×	×
	q_0	q_1	q_2	q_3

δ	a	b
(q_0, q_1)	(q_1, q_2)	(q_3, q_H)
(q_0, q_2)	(q_1, q_1)	(q_3, q_H)
(q_0, q_3)	(q_1, q_H)	(q_3, q_H)
(q_1, q_2)	(q_2, q_1)	(q_H, q_H)
(q_1, q_3)	(q_2, q_2)	(q_H, q_H)
(q_2, q_3)	(q_1, q_2)	(q_H, q_H)

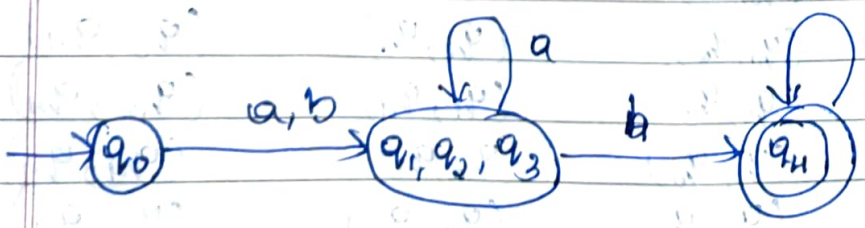
δ	a	b
(q_1, q_2)	(q_2, q_1)	(q_H, q_H)
(q_1, q_3)	(q_2, q_2)	(q_H, q_H)
(q_2, q_3)	(q_1, q_2)	(q_H, q_H)

The unmarked pairs (q_1, q_2), (q_1, q_3) and (q_2, q_3) are indistinguishable and the states q_0, q_H are distinguishable

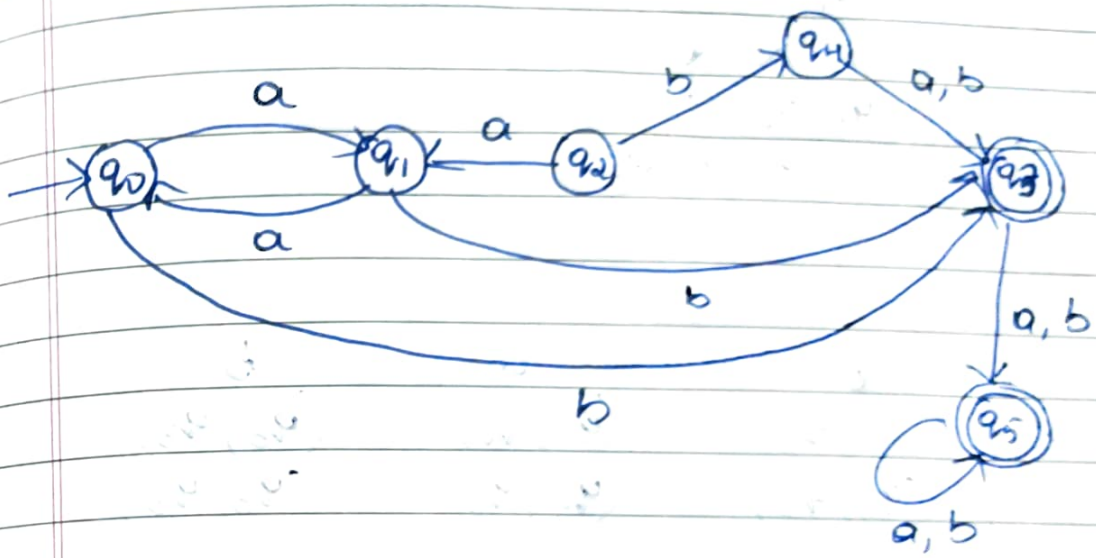
Indistinguishable group is q_1, q_2, q_3 and q_0, q_H are distinguishable group.

The various group that represents states of minimized DFA are
q_0, (q_1, q_2, q_3), q_H

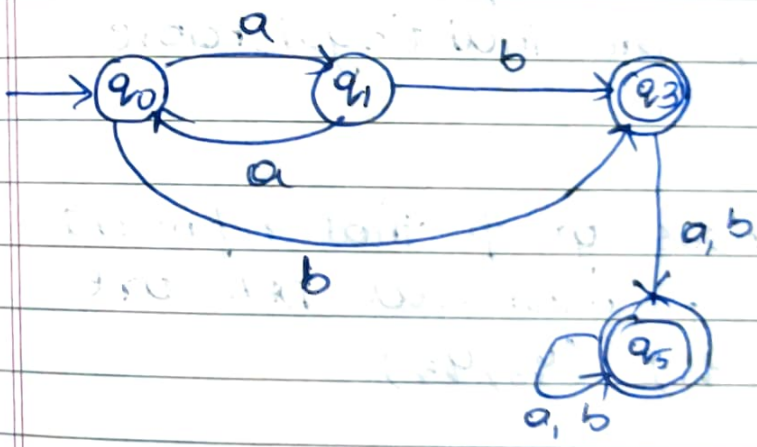
δ	a	b
→ q_0	(q_1, q_2, q_3)	(q_1, q_2, q_3)
(q_1, q_2, q_3)	(q_1, q_2, q_3)	q_H
* q_H	q_H	q_H



3] Minimize the following DFA.



⇒



Any state that is not reachable from start must be removed. (q_2, q_4)

(q_0, q_1) (q_0, q_3) (q_0, q_5)
(q_1, q_3) (q_1, q_5)
(q_3, q_5)

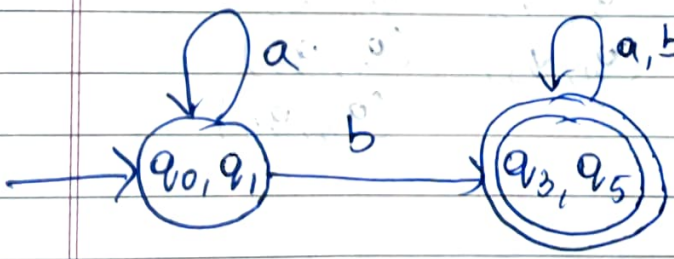
q_1			
q_3	X	X	
q_5	X	X	
	q_0	q_1	q_3

δ	a	b
(q_0, q_1)	(q_1, q_0)	(q_3, q_3)
(q_3, q_5)	(q_5, q_5)	(q_5, q_5)

The unmarked pairs are (q_0, q_1), (q_3, q_5) are indistinguishable

The various group that represents states of minimized DFA are
(q_0, q_1) (q_3, q_5)

δ	a	b
$\rightarrow (q_0, q_1)$	(q_0, q_1)	(q_3, q_5)
(q_3, q_5)	(q_3, q_5)	(q_3, q_5)



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Grammar:

A grammar G is a 4-tuple or quadruple, $G = (V, T, P, S)$ where

V → set of variables or non-terminals

T → is set of terminals

P → set of productions. Each production is of the form $\alpha \to \beta$ where α is a string in $(V \cup T)^+$ and hence α cannot be ε & ε cannot occur on the LHS of the production. but β is a string in $(V \cup T)^*$ and hence it includes ε also. So ε can occur on the RHS of the production.

S → start symbol.

Ex: $S \to AB$, $B \to bB$, $A \to a/\varepsilon$

$V = \{S, A, B\}$

$T = \{a, b, \varepsilon\}$

$P = \{S \to AB, B \to bB, A \to a, A \to \varepsilon\}$

S is the start symbol

Chomsky Hierarchy

Types of grammars

- Type 0 grammar (phrase structured grammar)
- Type 1 grammar (context sensitive grammar)
- Type 2 grammar (context free grammar)
- Type 3 grammar (Regular grammar)

Type 2 grammar (Context free grammar)

A grammar $G=(V, T, P, S)$ is said to be Type 2 grammar or context free grammar if all the productions are of the form $A \to \alpha$ then $\alpha \in (V \cup T)^*$ & and A is non-terminal.

Ex: $S \to \varepsilon$

$S \to aS$

$S \Rightarrow aS$
$\Rightarrow a\varepsilon$
$\Rightarrow a$

$S \Rightarrow aS$
$\Rightarrow aaS$
$\Rightarrow aaaS$
$\Rightarrow aaa\varepsilon$
$\Rightarrow aaa$

$L = \{\varepsilon, a, aa, aaa, \ldots\}$

$S \Rightarrow aS$
$\Rightarrow aaS$
$\Rightarrow aa\varepsilon$
$\Rightarrow aa$

$L = \{a^n, n \geq 0\}$

Grammar from Finite Automata.

Let $M = (Q, \Sigma, \delta, q_0, F)$ be a finite automata a accepting the language L where,

$Q = \{q_0, q_1, \ldots q_n\}$

$\Sigma = \{a_1, a_2, \ldots a_m\}$

A grammar $G = (V, T, P, S)$ can be constructed, where,

$V = \{q_0, q_1, \ldots q_n\}$ the states of DFA are the variables of grammar.

$T = \Sigma$ the input alphabets of DFA are the terminals in grammar

$S = q_0$ start state of DFA is the start symbol in grammar.

The productions P from the transitions can be obtained as follows:

i) $\delta(q_i, a) = q_j$ then introduce the production

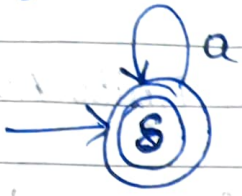
$$q_i \rightarrow aq_j$$

ii) If $q \in F$, then introduce the production

$$q \rightarrow \varepsilon$$

1. Obtain grammar to generate string consisting of any number of a's.

=



$S \to \varepsilon$

$\delta(S, a) = S$

$S \to aS$

P : { $S \to aS$, $S \to \varepsilon$ }

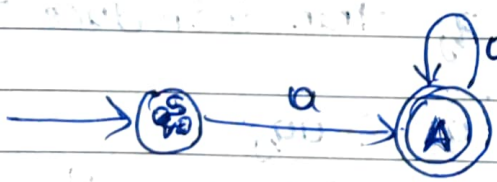
V = { S }

T = { a, ε }

S is the start symbol.

2. Obtain grammar to generate string consisting of atleast one 'a'.

=



V =

T =

S is the start symbol

$S \to a$

i) $\delta(S, a) = A$

$S \to aA$

ii) $\delta(A, a) = A$

$A \to aA$

P: { $S \to aA$, $A \to aA$, $A \to \varepsilon$ }

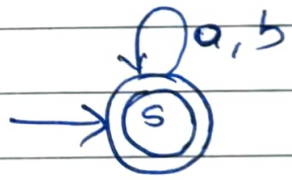
Minimum string from P.

$S \to aA$.

$S \to a\varepsilon$

$\to a$

3. Obtain grammar to generate string consisting of any number of a's & b's



$V = \{S\}$

$T = \{a, b, \varepsilon\}$

S is the start symbol

~~Sta~~ $\delta(S, a) = S$

$S \to aS$

$S \to a$S

$S \to abS$

$S \to ab$

$\delta(S, b) = S$

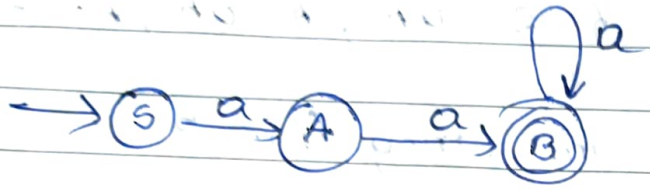
$S \to bS$

$S \to b$S

$S \to b$

P: { $S \to aS$, $S \to bS$, $S \to \varepsilon$ }

21. Obtain grammar to generate string consisting of atleast 2 numbers of a



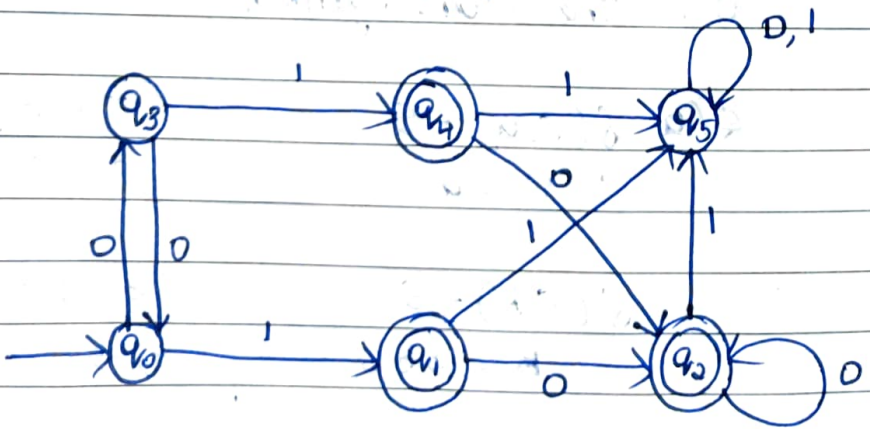
$\delta(S, a) = A$ $\quad S \to aA$

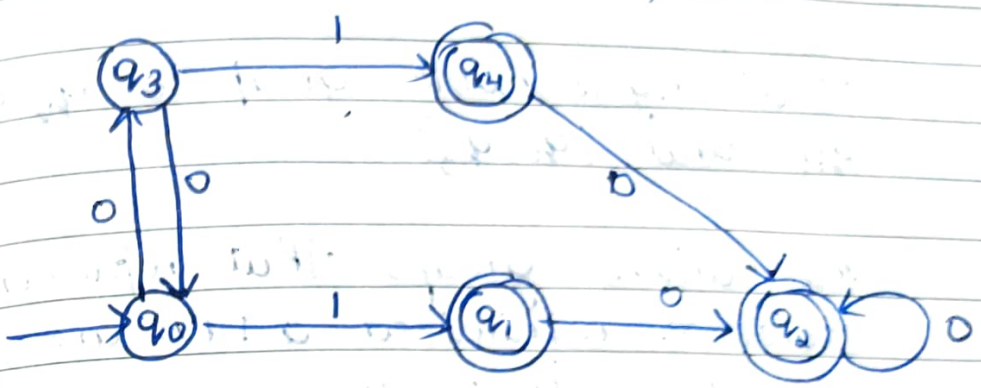
$\delta(A, a) = B$ $\quad A \to aB$

$\delta(B, a) = B$ $\quad B \to aB$

$B \to \varepsilon$ $\quad B \to \varepsilon$.

1. Identify distinguishable and indistinguishable states and minimize the DFA.





(q_0, q_1) (q_0, q_2) (q_0, q_3) (q_0, q_H)
(q_1, q_2) (q_1, q_3) (q_1, q_H)
(q_2, q_3) (q_2, q_H)
(q_3, q_H)

q_1	X			
q_2	X			
q_3		X	X	
q_H	X			X
	q_0	q_1	q_2	q_3

δ	0	1
(q_0, q_3)	(q_3, q_0)	(q_1, q_H)
(q_1, q_2)	(q_2, q_2)	ϕ
(q_1, q_H)	(q_2, q_2)	ϕ
(q_2, q_H)	(q_2, q_2)	ϕ

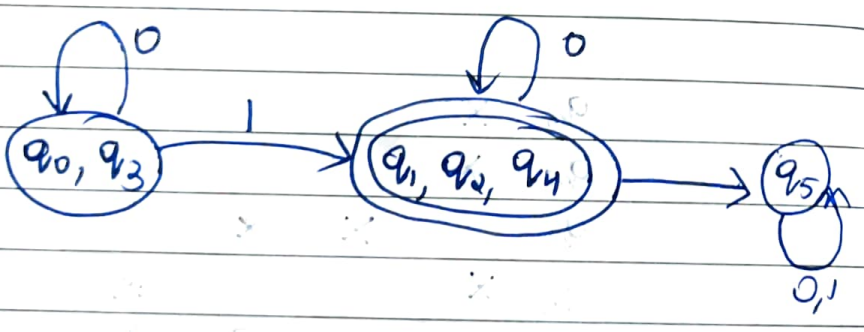
The unmarked pairs are (q_0, q_3), (q_1, q_2), (q_1, q_H) and (q_2, q_H) are

indistinguishable

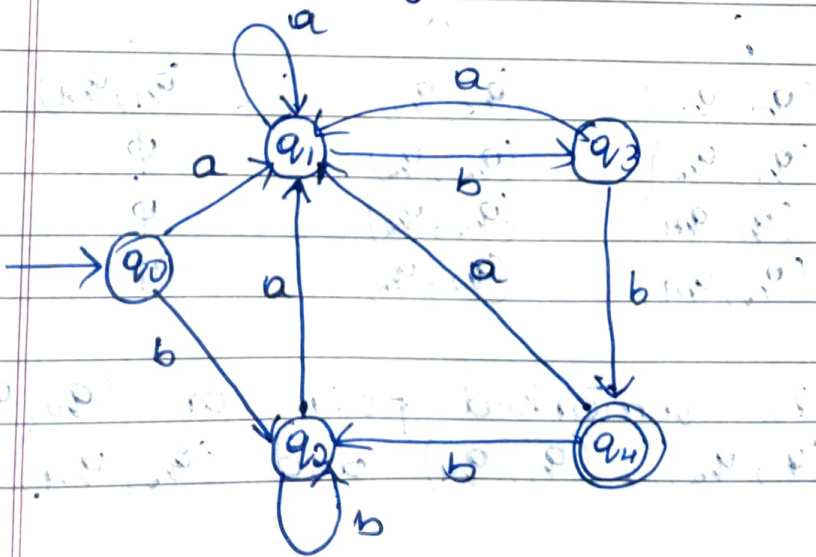
Indistinguishable group is q_0 q_1, q_2, q_4 and (q_0, q_3)

The various groups that represent states of minimized a DFA are
(q_0, q_3), (q_1, q_2, q_4)

δ	0	1
$\rightarrow(q_0, q_3)$	(q_0, q_3)	(q_1, q_2, q_4)
$*(q_1, q_2, q_4)$	(q_1, q_2, q_4)	ϕ



2. Minimize the given DFA



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(q_0, q_1) (q_0, q_2) (q_0, q_3) (q_0, q_H)
(q_1, q_2) (q_1, q_3) (q_1, q_H)
(q_2, q_3) (q_2, q_H)
(q_3, q_H)

q_1	X			
q_2		X		
q_3	X	X	X	
q_H	X	X	X	X
	q_0	q_1	q_2	q_3

δ	a	b
(q_0, q_1)	(q_1, q_1)	(q_2, q_3)
(q_0, q_2)	(q_1, q_1)	(q_2, q_2)
$-(q_0, q_3)$	(q_1, q_1)	(q_2, q_H)
(q_1, q_2)	(q_1, q_1)	(q_3, q_2)
$-(q_1, q_3)$	(q_1, q_1)	(q_3, q_H)
$-(q_2, q_3)$	(q_1, q_1)	(q_2, q_H)

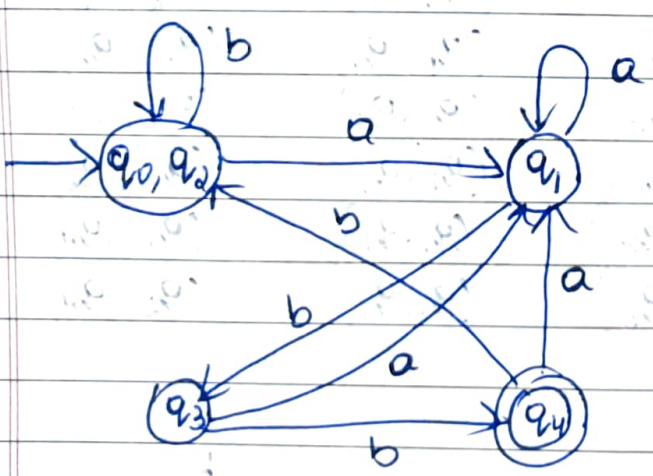
δ	a	b
(q_0, q_1)	(q_1, q_1)	(q_2, q_3)
(q_0, q_2)	(q_1, q_1)	(q_2, q_2)
$-(q_1, q_2)$	(q_1, q_1)	(q_2, q_2)

δ	a	b
(q_0, q_2)	(q_1, q_1)	(q_2, q_2)
$(\cancel{q_1, q_2})$	$(\cancel{q_1, q_1})$	

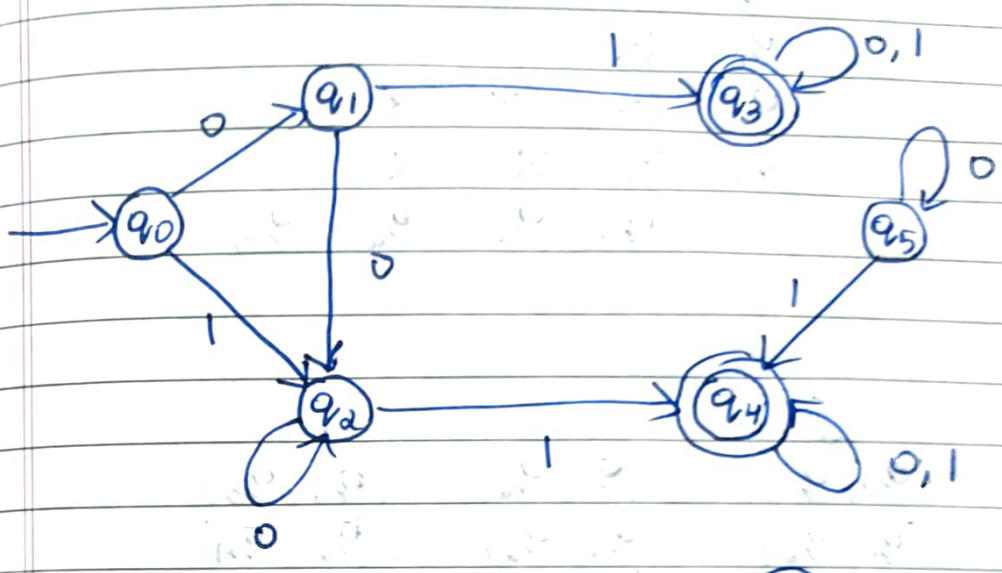
The unmarked pairs are (q_0, q_2) are indistinguishable and q_1, q_3, q_4 are distinguishable

The various group that represents states of minimized DFA are (q_0, q_2), q_1, q_3, q_4.

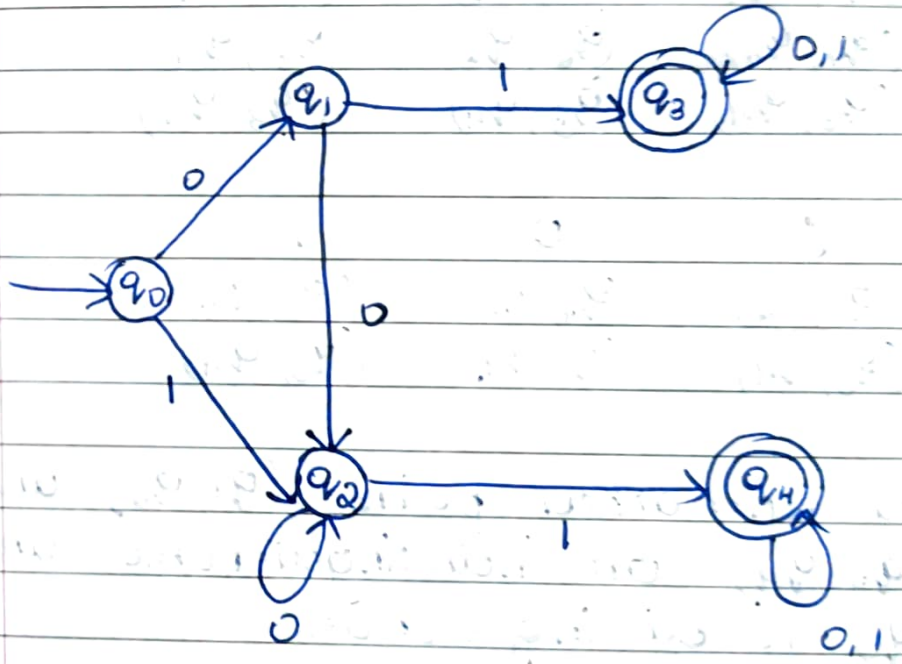
δ	a	b
$\rightarrow (q_0, q_2)$	q_1	(q_0, q_2)
q_1	q_1	q_3
q_3	q_1	q_4
$* q_4$	q_1	(q_0, q_2)



1) Minimize the following DFA:



=



= (q_0, q_1) (q_0, q_2) (q_0, q_3) (q_0, q_H)
(q_1, q_2) (q_1, q_3) (q_1, q_H)
(q_2, q_3) (q_2, q_H)
(q_3, q_H)

q_1	X			
q_2	X			
q_3	X	X	X	
q_4	X	X	X	
	q_0	q_1	q_2	q_3

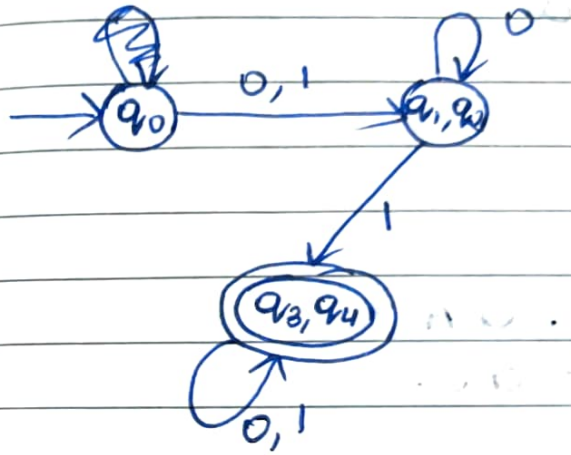
δ	0	1
(q_0, q_1)	(q_1, q_2)	(q_2, q_3)
(q_0, q_2)	(q_1, q_2)	(q_2, q_4)
(q_1, q_2)	(q_2, q_2)	(q_3, q_4)
(q_3, q_4)	(q_3, q_4)	(q_3, q_4)

δ	0	1
(q_1, q_2)	(q_2, q_2)	(q_3, q_4)
(q_3, q_4)	(q_3, q_4)	(q_3, q_4)

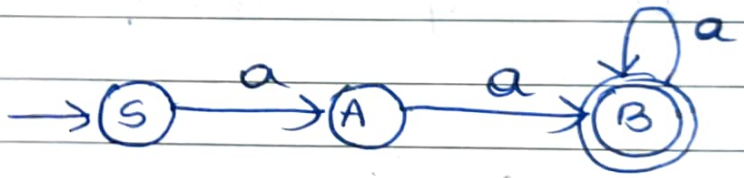
The unmarked pairs (q_1, q_2) and (q_3, q_4) are indistinguishable and q_0 is distinguishable.

The various group that represents states of minimized DFA are (q_1, q_2), (q_3, q_4), q_0.

δ	0	1
$\rightarrow q_0$	(q_1, q_2)	(q_1, q_2)
(q_1, q_2)	(q_1, q_2)	(q_3, q_4)
$*(q_3, q_4)$	(q_3, q_4)	(q_3, q_4)



2. Obtain the grammar to generate a string consisting of atleast 2 a's



$V = \{S\}$

$T = \{a, \varepsilon\}$

$\delta(S, a) = aA$

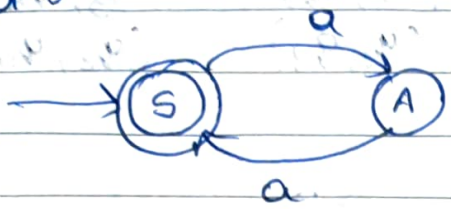
$\delta(A, a) = aB$

$\delta(B, a) = aB$

$P = \{ S \rightarrow aA,\ A \rightarrow aB,\ B \rightarrow aB,\ B \rightarrow \varepsilon \}$

3. Obtain a grammar to generate a string consisting of even numbers of a's

:



$V = \{S, A\}$

$T = \{a, \varepsilon\}$

$\delta(S, a) = aA$

$\delta(A, a) = aS.$

1) Obtain a grammar to generate a string consisting of multiples of 3 a's.

=

$S \rightarrow aaaS/\varepsilon$

$\varepsilon \quad S \Rightarrow aaaS$

aaa

$\{\varepsilon, aaa, aaaaaa, \ldots\}$

$S \Rightarrow aaaS$

$\Rightarrow aaaaaaS$

$\Rightarrow aaaaaa (S \rightarrow \varepsilon)$

a) obtain a grammar to generate string's of a's and b's such that string length is multiple of 3.

$S \rightarrow AAAS / \varepsilon$

$A \rightarrow a / b$

$S \Rightarrow AAAS \quad (A \rightarrow a)$

$\Rightarrow aAAS \quad (A \rightarrow b)$

$\Rightarrow abAS \quad (A \rightarrow b)$

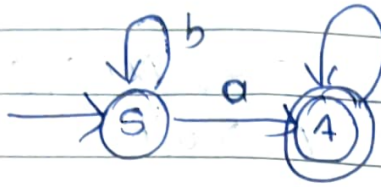
$\Rightarrow abbS \quad (S \rightarrow \varepsilon)$

$\Rightarrow abb$

3) Dervie a string of length 6.

$S \rightarrow AAAA$

4) Obtain a grammar to generate string consisting of any numbers of a's and b's with atleast one a's



= $S \to a$
$S \to aS$
$S \to bS$

$\delta(S, b) = S \Rightarrow S \to bS$
$\delta(S, b) = A \quad S \to aS$
$\delta(A, a) = A \quad A \to aA$
$\delta(A, b) = A \quad A \to bA$
$A \to \varepsilon$

$S \Rightarrow aS$
$\Rightarrow abS$
$\Rightarrow aba$

$S \Rightarrow bS$
$\Rightarrow ba$

$S \Rightarrow bS \quad (S \to bS)$
$\Rightarrow bbS \quad (S \to bS)$
$\Rightarrow bbbS \quad (S \to bS)$
$\Rightarrow bbba \quad (S \to a)$

5) obtain a grammar to generate string consisting of any numbers of a's and b's with atleast one b

$\Rightarrow$ $S \to b$
$S \to aS$
$S \to bS$

6) Obtain a grammar to generate string consisting of any number of a's and b's with atleast one a or b

=
$S \to b/a$
$S \to aS$
$S \to bS$

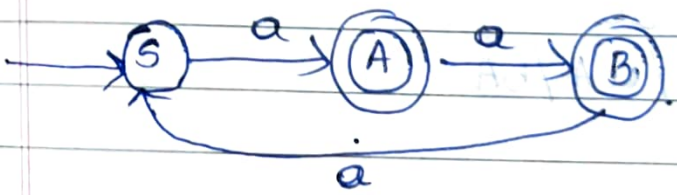
imp
7) Obtain a grammar to accept the following language
$L = \{w : |w| \bmod 3 > 0$ where w ∈ ∑*

=
$S \to a/aa$
$S \to aaaS$

$S \Rightarrow aaaS$
$\Rightarrow aaaa$ $(S \to a)$

$S \Rightarrow aaaS$
$\Rightarrow aaaaaaS$
$\Rightarrow aaaaaaaa$ $(S \to a)$

$S \Rightarrow aaaS$
$aaaaa$ $(S \to aa)$



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Grammar From Regular Expression

1. Obtain grammar to generate strings of a's and b's having a substring ab.

= R·E = $(a+b)^*$ ab $(a+b)^*$

A ab A

$S \rightarrow AabA$
$A \rightarrow \varepsilon / aA / bA$.

2. Obtain gramar to generate strings of a's and b's ending with string ab

= R·E = $(a+b)^*$ ab

$S \rightarrow Aab$
$A \rightarrow \varepsilon / aA / bA$

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Derivation:

The process of obtaining strings of terminals and/or non-terminals from the start symbol by applying some or all productions is called derivation.

Let $A \to \alpha B \gamma$ and $B \to \beta$ are two production in grammar G, where α, β, and γ are strings of terminals and/or non-terminals, A and B are non-terminals.

A produces the string $\alpha\beta\gamma$ by replacing B in $\alpha B \gamma$ by $B \to \beta$.

Ex: $E \to E+E$
$E \to E-E$
$E \to E*E$
$E \to E/E$
$E \to id$

Derive id + id * id

Left Most Derivation

$E \Rightarrow E+E \quad (E \to E+E)$
$\Rightarrow id+E \quad (E \to id)$
$\Rightarrow id+E*E \quad (E \to E*E)$
$\Rightarrow id+id*E \quad (E \to id)$
$\Rightarrow id+id*id \quad (E \to id).$

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$E \Rightarrow E * E$ $(E \to E * E)$
$E * id$ $(E \to id)$
$E + E * id$ $(E \to E + E)$
$E + id * id$ $(E \to id)$
$id + id * id$ $(E \to id)$

} sentential forms.

$E \overset{*}{\underset{rm}{\Longrightarrow}} id + id * id$
sentence of string

Left Most Derivation: (LMD)

In the derivation process if a left most variable is replaced at every step then the derivation is said to be Left most.

Right Most Derivation: (RMD)

In the derivation process if a right most variable is replaced at every step then the derivation is said to be Right Most.

1. Obtain the LMD for the string v aaabbabbba using the following grammar

$S \to aB / bA$
$A \to aS / bAA / a$
$B \to bS / aBB / b$

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$S \Rightarrow aB$ $(S \to aB)$

$aaBB$ $(B \to aBB)$

$aaaBBB$ $(B \to aBB)$

$aaabBB$ $(B \to b)$

$aaabbB$ $(B \to b)$

$aaabbaBB$ $(B \to aBB)$

$aaabbabB$ $(B \to b)$

$aaabbabbS$ $(B \to bS)$

$aaabbabbbA$ $(B \to S \to bA)$

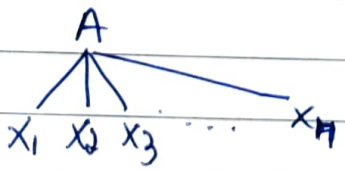
$aaabbabbba$ $(A \to a)$

$S \overset{*}{\underset{LM}{\Longrightarrow}} aaabbabbba$

Derivation Tree or Parse Tree

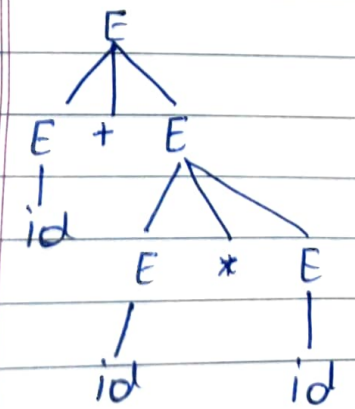
Let $G = (V, T, P, S)$ be a CFG. The tree is derivation tree or Parse Tree with the following properties:

1. The root has the label S.
2. Every vertex has a label which is a $(V \cup T \cup \varepsilon)$ (union)
3. Every leaf node has label from T and an interior vertex has a label from V.
4. If a vertex is labeled A and if $X_1 X_2 X_3 \cdots X_n$ are all children of A from left, then
   $A \rightarrow X_1 X_2 X_3 \cdots X_n$ must be a production in P

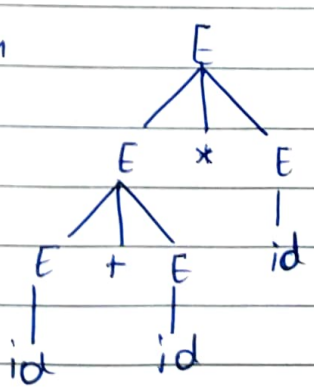


Ex: id + id * id

LM



RM



$\Rightarrow_{lm}$ E + E

E + E * E

E + E * id

E + id * id

id + id * id

Ambigous Grammar:

Let G = (V, T, P, S) be a CFG. A grammar G is a ambigous iff there exists atleast one string w belongs to T^* ($w \in T^*$) for which two or more different Parse Tree consists by applying either a LMD or RMD.

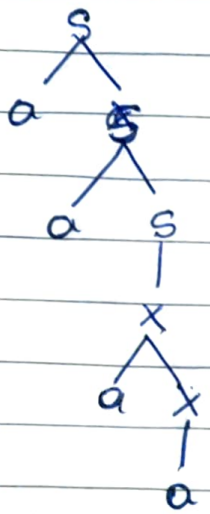
1. Is the following grammar ambigous.

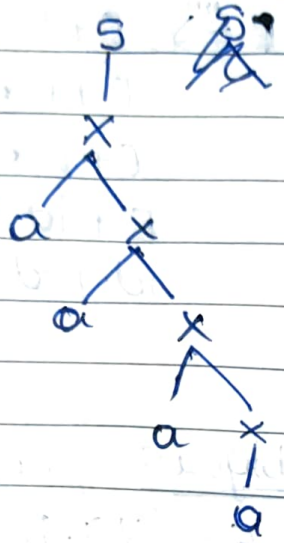
S → aS / X

X → aX / a

=

S ⇒ aS	S ⇒ X
aaS	⇒ aX
aaX	⇒ aaX
aaaX	⇒ aaaX
aaaa	⇒ aaaa





2. Is the following grammar ambigous

$S \rightarrow aB / bA$

$A \rightarrow aS / bAA / a$

$B \rightarrow bS / aBB / b$

1. obtain a grammar to generate the following language.

i) $L = \{a^n b^n : n \geq 0\}$

ii) $L = \{a^n b^n : n \geq 1\}$

i) = $S \to \varepsilon / asb$

$S \Rightarrow asb$
$a\varepsilon b$
ab

$S \Rightarrow asb$
$\Rightarrow aasbb$
$\Rightarrow aabb$

ii) $S \to ab / asb$

$S \Rightarrow asb$
$\Rightarrow aabb$

$S \Rightarrow asb$
$\Rightarrow aasbb$
$\Rightarrow aaabbb$

iii) $L = \{a^{n+1} b^n : n \geq 0\}$

= $S \to a / asb$

$S \Rightarrow asb$
$\Rightarrow aab$

$S \Rightarrow asb$
$\Rightarrow aasbb$
$\Rightarrow aaabb$

iv) $L = \{a^n b^{n+1} : n \geq 0\}$

$S \to b / aSb$

$V = \{S\}$
$T = \{a, b\}$
$P = \{S \to b / asb\}$
S is the start state

v) $L = \{a^n b^{n+2} : n \geq 0\}$

= $S \to bb / asb$

$V = \{S\}$
$T = \{a, b\}$
$P = \{ S \to bb / asb \}$
S is start symbol

6. $L = \{a^n b^{2n} : n \geq 0\}$

= $S \to \varepsilon / aSbb$

$S \Rightarrow aSbb$
$\Rightarrow abb$

$S \Rightarrow asbb$
$\Rightarrow aaSbbbb$
$\Rightarrow aaBbbb$

$V = \{S\}$
$T = \{a, b\}$
$P = \{ S \to \varepsilon / asbb \}$
S is start symbol

7. Let $\Sigma = \{a, b\}$. Obtain grammar G generating set of all palindromes over Σ

= $S \to \varepsilon$
$S \to a / b$
$S \to aSa / bSb$

$S \Rightarrow bSb$
$\Rightarrow baSab$
$\Rightarrow baaab$

8. $L = \{ww^R$ where $w \in \{a,b\}^* \}$

$S \to asa / bsb$

$S \to \varepsilon$

$S \to asa$

$\Rightarrow abSba$

$\Rightarrow abasaba$

$\Rightarrow ababaaba$

9. $L = \{wcw^R / w \in \{a,b\}^* \}$

$S \to asa / bsb$

$S \to c$

1. Obtain a grammar to generate a language consisting of all non-palindrome over set $\{a,b\}$

2. $L = \{0^m, 1^m, 2^n / m \geq 1 \text{ and } n \geq 0\}$

3. $L = \{w / n_a(w) = n_b(w)\}$ $\Sigma = \{a,b\}$

4. What is the language genirated by the grammar

$S \to 0A / \varepsilon$

$A \to 1S$

=

$S \Rightarrow 0A$
$\Rightarrow 01S$
$\Rightarrow 01$

$S \Rightarrow 0A$
$\Rightarrow 01S$
$\Rightarrow 010A$
$\Rightarrow 0101S$
$\Rightarrow 0101$

$L = \{(01)^n : n \geq 0\}$

$L = \{w / w \in (01)^*\}$

2. What is the language generated by the grammar

$S \rightarrow aCa$;
$C \rightarrow aCa / b$

=

$S \Rightarrow aCa$
$\Rightarrow abA$

$L = \{a^n b a^n : n \geq 1\}$

3. Obtain a CFG to generate a string of balanced paranthesis.

=

$S \rightarrow (S)$
$S \rightarrow [S]$
$S \rightarrow \{S\}$
$S \rightarrow SS$
$S \rightarrow \varepsilon$

Ex: $([])\{\}$
$\{[](){}\}$

$S \Rightarrow SS$
$\Rightarrow (S)S$
$\Rightarrow ([S])S$
$\rightarrow ([\,])S$
$\Rightarrow ([\,])\{S\}$
$\Rightarrow ([\,])\{\,\}$
$=$

~~Obtain a string aaabbabbba~~

1. Is the following ambigous

$S \rightarrow iCtS / iCtSeS / a$
$C \rightarrow b$

1. Obtain the unambigous grammar

$E \to E+E / E-E$

$E \to E*E / E/E$

$E \to (E) / I$

$I \to a/b/c$

highest

$=$ $E \to E+T / E-T / T$

$T \to T*F / T/F / F$

$F \to (E) / I$

$I \to a/b/c$

$(a+b)*(a-b)$ → Derivie

~~$T \to T*F$~~

$E \Rightarrow T$

$\Rightarrow T*F$

$F*F$

$(E)*F$

$(E+T)*F$

~~$(E+T)*(E)$~~

~~$(E+T)*(E-T)$~~

$(T+T)*F$

$(F+T)*F$

$(I+T)*F$

$(a+T)*F$

$(a+F)*F$

$(a+I)*F$

$(a+b)*F$

$(a+b)*(E)$

(a+b) * (E-T)
(a+b) * (T-T)
(a+b) * (F-T)
(a+b) * (I-T)
(a+b) * (a-T)
(a+b) * (a-F)
(a+b) * (a-I)
(a+b) * (a-b)

i) If we assume + & - are having highest component to *, / all are left associated then grammar

E → E*E / E/E
E → E+E / E-E
E → (E) / I
I → a/b/c

⇒ E → E*T / E/T / T
T → T*F / T-F / F
F → (E) / I
I → a/b/c

ii) If we assume + & - are having highest priority than *, / and all the operats are right associated

E → E*E / E/E
E → E+E / E-E

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$E \rightarrow (E) / I$

$I \rightarrow a/b/c$

$E \rightarrow T * E / T / E / T$

$T \rightarrow F + T / F - T / F$

$F \rightarrow (E) / I$

$I \rightarrow a/b/c$

iii) If we assume + & - having highest priority and left associative when compared to that of / & *. and if * & / operators are right associative

⇒ $E \rightarrow E * E / E / E$

$E \rightarrow E + E / E - E$

$E \rightarrow (E) / I$

$I \rightarrow a/b/c$

= $E \rightarrow T * E / T / E / T$

$T \rightarrow T + F / T - F / F$

$F \rightarrow (E) / I$

$I \rightarrow a/b/c.$

Inherently Ambigous Grammar

If there exist a language L for which there is no unambigous grammar then the language is called inherently ambigous ~~grammar~~ language and the grammar from which language is dervied is called inherently ambigous grammar.

co-

Push Down Automata (PDA)

A push down automata(PDA) is a seven-tuple:

$$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$$

where,

M → name of the machine

Q → non-empty finite set of states

Σ → non-empty finite set of input alphabet

Γ → set of stack alphabets

δ → is a transition function from $Q \times (\Sigma \cup \varepsilon) \times \Gamma$ to finite subset of $Q \times \Gamma^*$

q_0 → $q_0 \in Q$ is the start state

z_0 → $z_0 \in \Gamma$ is the initial symbol on the stack.

F → $F \subseteq Q$ is set of final state

The transition performed by PDA depends on

* current state
* current input symbol
* symbol on top of stack

The actions performed by machine consist of changing states from one state to another replacing symbol on the stack.

* δ(state, input symbol, stack symbol) = (state, stack symbol)

1). What does each of the following transition represent?

a) $\delta(P, a, z) = (q, az)$

$Q \times \Sigma \cup \epsilon \times \Gamma \quad Q \times \Gamma^*$

b) $\delta(P, a, z) = (q, \epsilon)$ $\Rightarrow$ z is popped from stack

c) $\delta(P, a, z) = (q, r)$ $\Rightarrow$ z is popped and replaced by r

d) $\delta(P, \epsilon, z) = (q, r)$

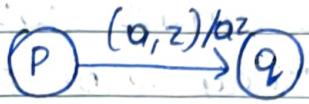
e) $\delta(P, \epsilon, \epsilon) = (q, z)$

f) $\delta(P, \epsilon, z) = (q, \epsilon)$

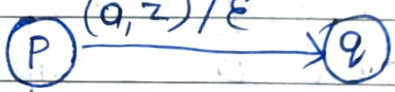
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Graphical Representation of PDA:

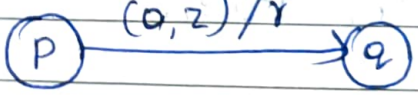
1. $\delta(p, a, z) = (q, az)$

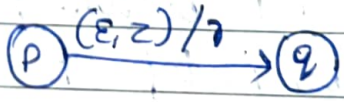


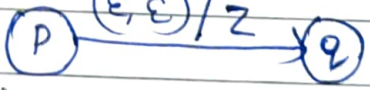
2. $\delta(P, a, z) = (q, \varepsilon)$

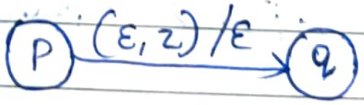


3. $\delta(P, a, z) = (q, r)$



d 

e 

f 

Instantaneous Description (ID)

The current configuration of PDA at any given instant can be described by an instantaneous description (ID). An ID gives the current state of PDA, the remaining string to be processed and the entire contents of the stack.

Let $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$ be a PDA an ID is defined as 3-tuple or triple (q, w, α) where q is a current state, w is a string to be processed and α is the current contents of the stack.

[Right-most symbol is bottom of stack
Left most symbol is top of stack]

$(q, aw, z\alpha)$

$\delta(q, a, z) = (p, \beta) \rightarrow (p, w, \beta\alpha)$

$(q, aw, z\alpha) \vdash (p, w, \beta\alpha)$

↓ ID representation

Acceptance of Language by PDA:

There are two cases where in a string w is accepted by PDA:

i) get the final state from start state
ii) get an empty stack from start state

Get the final state from start state

Let $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$ be a PDA the language $L(M)$ accepted by a final state is defined as

$$L(M) = \{ w / (q_0, w, z_0) \vdash^* (p, \epsilon, \alpha), p \in F \}$$

Get an Empty stack from start state

Let $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$ be a PDA the language $L(M)$ accepted by a final state is defined as

$$L(M) = \{ w / (q_0, w, z_0) \vdash^* (p, \epsilon, \epsilon) \}, p \in Q$$

Construction of PDA

1. Obtain a PDA to accept the language $L(M) = \{ w c w^R / w \in (a+b)^* \}$, where w^R is reverse of w by a final state

- To check for the palindrome let us push all scanned symbols on to stack till we encounter the letter c

Once we pass the middle string if the string is palindrome, for each scanned input symbol there should be corresponding symbol (same as input symbol) on the stack.

Finally if there is no input and stack is empty we say that the given string is palindrome.

Step1: $\delta(q_0, a, z_0) = (q_0, az_0)$
$\delta(q_0, b, z_0) = (q_0, bz_0)$

Let q_0 be the initial state and z_0 be the initial symbol on the stack. In state q_0 and where the top of the stack is z_0, whether the input symbol is a or b push it on to the stack and remain on q_0

Step2: $\delta(q_0, a, a) = (q_0, aa)$
$\delta(q_0, a, b) = (q_0, ab)$

$\delta(q_0, b, a) = (q_0, ba)$
$\delta(q_0, b, b) = (q_0, bb)$

Once the first scanned input symbol is pushed on to the stack, the stack

baaG

may contain either a or b.
Now in state q_0 the input symbol can be either a or b.
Note that irrespective of what is the input or what is there on the stack we have to keep pushing all the symbols on to the stack till we encounter C.

step2: $\left\{ \begin{array}{l} \delta(q_0, C, z_0) \\ \delta(q_0, C, a) \\ \delta(q_0, C, b) \end{array} \right\}$

If the next input symbol is C, the top of the stack either a or b, another possibility is that in state q_0 the first input symbol itself can be C.
In this case w is null string and z_0 will be on the stack.
In all these cases the input symbol is C that is the symbol which is present in the middle of the string. So, change the state to q_1 and donot alter the contents of the stack and the transition for these will be

$$\delta(q_0, C, z_0) = (q_1, z_0)$$
$$\delta(q_0, C, a) = (q_1, a)$$
$$\delta(q_0, C, b) = (q_1, b)$$

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Step 3:
Input symbols can be either a or b to be a palindrome for each input symbol there should be corresponding symbol on the stack. So, whenever the input symbol is same as the symbol on stack remain in state q_1 and delete the symbol from the stack and repeat the process

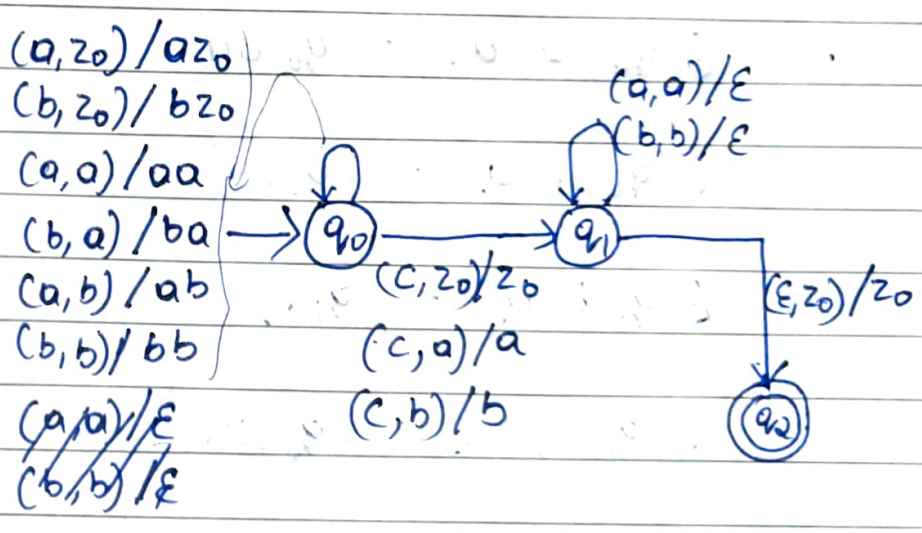
$$\delta(q_1, a, a) = (q_1, \varepsilon)$$
$$\delta(q_1, b, b) = (q_1, \varepsilon)$$

Step 4:
Finally in state q_1 if the string is palindrome there is no input symbol to be scanned and the stack should be empty (the stack should contain z_0) Now, change the state to q_2 and donot alter the contents of the stack. and the transition for this will be

$$\delta(q_1, \varepsilon, z_0) = (q_2, z_0)$$

PDA : $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$
where $Q = \{q_0, q_1, q_2\}$
$\Sigma = \{a, b, c\}$
$\Gamma = \{a, b, z_0\}$
δ :

$\delta(q_0, a, z_0) = (q_0, az_0)$
$\delta(q_0, b, z_0) = (q_0, bz_0)$
$\delta(q_0, a, a) = (q_0, aa)$
$\delta(q_0, b, a) = (q_0, ba)$
$\delta(q_0, a, b) = (q_0, ab)$
$\delta(q_0, b, b) = (q_0, bb)$
$\delta(q_0, c, z_0) = (q_1, z_0)$
$\delta(q_0, c, a) = (q_1, a)$
$\delta(q_0, c, b) = (q_1, b)$
$\delta(q_1, a, a) = (q_1, \varepsilon)$
$\delta(q_1, b, b) = (q_1, \varepsilon)$
$\delta(q_1, \varepsilon, z_0) = (q_2, z_0)$



$q_0 \in Q$ is the start state of the machine
$z_0 \in \Gamma$ is the initial symbol on the stack
$F = \{q_0\}$ is the final state.

Ex: aabCbaa

$(q_0, aabCbaa, z_0) \vdash (q_0, abCbaa, az_0)$
$\vdash (q_0, bCbaa, aaz_0)$

$\vdash (q_0, cbaa, baaz_0)$
$\vdash (q_{01}, baa, baaz_0)$
$\vdash (q_1, aa, aaz_0)$
$\vdash (q_1, a, az_0)$
$\vdash (q_1, \varepsilon, z_0)$
$\vdash (q_2, \varepsilon, z_0)$

2. Obtain a PDA to accept the language $L(w) = \{a^n b^n, n \geq 1\}$ by a final state

$\delta(q_0, a, z_0) = (q_0, az_0)$

$(n-1)$ $\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, a) = (q_1, \varepsilon)$

$(n-1)$ $\delta(q_1, b, a) = (q_1, \varepsilon)$

by empty stack

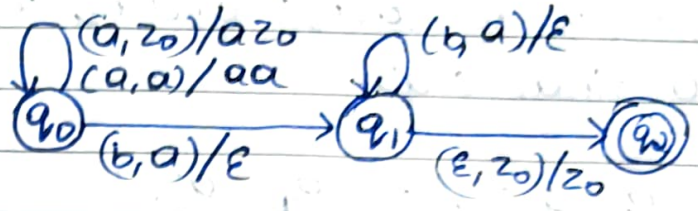
$\delta(q_1, \varepsilon, z_0) = (q_2, z_0) \Rightarrow \delta(q_1, \varepsilon, z_0) = (q_1, z_0)$

PDA: $M = (Q, \Sigma, \delta, \Gamma, q_0, z_0, F)$

$Q = \{q_0, q_1, q_2\}$
$\Sigma = \{a, b\}$
δ :

$\delta(q_0, a, z_0) = (q_0, az_0)$
$\delta(q_0, a, a) = (q_0, aa)$
$\delta(q_0, b, a) = (q_1, \varepsilon)$
$\delta(q_1, b, a) = (q_1, \varepsilon)$ $\delta(q_1, \varepsilon, z_0) = (q_2, z_0)$

$\Gamma = \{a, b z_0\}$



$q_0 \in Q$ is the start state of the machine
$q_2 \in F$ is the final state
$z_0 \in \Gamma$ is the initial symbol on the stack

Show the ID for the given string
aaabbb

$(q_0, aaabbb, z_0) \vdash (q_0, aabbb, az_0)$
$\vdash (q_0, abbb, aaz_0)$
$\vdash (q_0, bbb, aaaz_0)$
$\vdash (q_1, bb, aaz_0)$
$\vdash (q_1, b, az_0)$
$\vdash (q_1, \varepsilon, z_0)$
$\vdash (q_2, \varepsilon, z_0)$
$\Rightarrow$ final configuration

aab
$(q_0, aab, z_0) \vdash (q_0, ab, az_0)$
$\vdash (q_0, b, aaz_0)$
$\vdash (q_1, \varepsilon, az_0)$
$\therefore$ Not defined

3. Obtain a PDA to accept the language $L(w) = \{w / w \in (a+b)^* \text{ and } n_a(w) = n_b(w)\}$ by a final state

$$\delta(q_0, a, z_0) = (q_0, az_0)$$
$$\delta(q_0, b, z_0) = (q_0, bz_0)$$

$$\delta(q_0, a, a) = (q_0, aa)$$
$$\delta(q_0, a, b) = (q_0, \varepsilon)$$

$$\delta(q_0, b, a) = (q_0, \varepsilon)$$
$$\delta(q_0, b, b) = (q_0, bb)$$

by empty stack.
but q_1 is not final state

$$\delta(q_0, \varepsilon, z_0) = (q_1, z_0) \Rightarrow \delta(q_0, \varepsilon, z_0) = (q_1, z_0)$$

$q_0 \rightarrow q_1$

PDA: $M = (Q, \Sigma, \delta, \Gamma, q_0, z_0, F)$

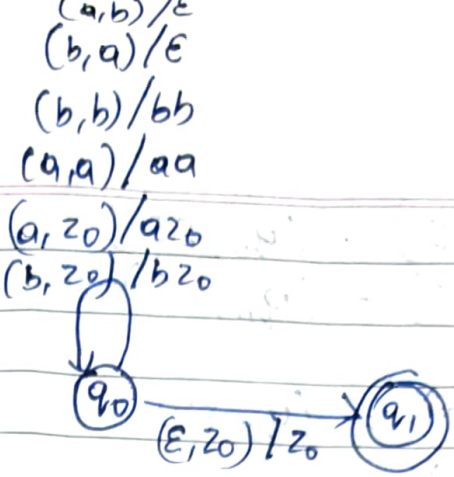
$Q = \{q_0, q_1\}$

$\Sigma = \{a, b\}$

δ:

$$\delta(q_0, a, z_0) = (q_0, az_0)$$
$$\delta(q_0, b, z_0) = (q_0, bz_0)$$
$$\delta(q_0, a, a) = (q_0, aa)$$
$$\delta(q_0, a, b) = (q_0, \varepsilon)$$
$$\delta(q_0, b, a) = (q_0, \varepsilon)$$
$$\delta(q_0, b, b) = (q_0, bb)$$
$$\delta(q_0, \varepsilon, z_0) = (q_1, z_0)$$

$\Gamma = \{a, b, z_0\}$



$q_0 \in Q$ is the start symbol on the stack

$z_0 \in \Gamma$ is the initial symbol on stack

$F = \{q_1\}$ is the final state.

ab aabb.

$(q_0, abaabb, z_0) \vdash (q_0, baabb, az_0)$
$\vdash (q_0, aabb, z_0)$
$\vdash (q_0, abb, az_0)$
$\vdash (q_0, bb, aaz_0)$
$\vdash (q_0, b, az_0)$
$\vdash (q_0, \varepsilon, z_0)$
$\vdash (q_1, z_0)$

imp

4. Obtain a PDA to accept a string of balanced parentheses. The parenthesis to be considered are (), [,]

Ex: 1) [()()([])]

2) ε

3) [][]()

step: $\delta(q_0, (, z_0) = (q_0, (z_0)$

$\delta(q_0, [, z_0) = (q_0, [z_0)$

step2 $\delta(q_0, (, () = (q_0, ((), \quad \delta(q_0, (, [) = (q_0, ([)$

$\delta(q_0, [, [) = (q_0, [[)$

$\delta(q_0, [, ( ) = (q_0, [( )$

$\delta(q_0,), ( ) = (q_0, \epsilon)$

~~$\delta(q_0,), [)$ =~~

~~$\delta(q_0,], ( )$ =~~

~~$\delta(q_0,], [)$ =~~

$\delta(q_0,], [) = (q_0, \epsilon)$

steps:

$\delta(q_0, \epsilon, z_0) = (q_1, z_0)$

PDA: $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$

$Q = \{q_0, q_1\}$

$\Sigma = \{(,), [,]\}$

$\Gamma = \{(, [, z_0\}$

δ:

$\delta(q_0, (, z_0) = (q_1, (z_0)$

$\delta(q_0, [, z_0) = (q, [z_0)$

$\delta(q_0, (, () = (q, ((\,)$

$\delta(q_0, (, [) = (q, ([)$

$\delta(q_0, [, () = (q, [(\,)$

$\delta(q_0, [, [) = (q, [[)$

$\delta(q_0,), () = (q, \epsilon)$

$\delta(q_0,], [) = (q, \epsilon)$

$\delta(q_0, \epsilon, z_0) = (q, z_0)$

$L(w) = ww^R / w \in (a+b)^*$

$L(w) = \{w / w \in (a+b)^* \; n_a(w) > n_b(w)$

5. Obtain PDA to accept the language $L(w) = \{a^n b^{2n}, n \geq 1\}$ by a final state

1. Obtain a PDA to accept the language $L(w) = \{wcw^R / w \in (a+b)^*\}$ where w^R is reverse of w by a empty stack

=

$\delta(q_0, a, z_0) = (q_0, az_0)$
$\delta(q_0, b, z_0) = (q_0, bz_0)$
$\delta(q_0, a, a) = (q_0, aa)$
$\delta(q_0, a, b) = (q_0, ab)$
$\delta(q_0, b, a) = (q_0, ba)$
$\delta(q_0, b, b) = (q_0, bb)$
$\delta(q_0, c, z_0) = (q_1, z_0)$
~~$\delta(q_1, \varepsilon, z_0) = (q_1, \varepsilon)$~~
$\delta(q_0, c, a) = (q_1, a)$
$\delta(q_0, c, b) = (q_1, b)$
$\delta(q_1, 0, a) = (q_1, \varepsilon)$
$\delta(q_1, b, b) = (q_1, \varepsilon)$
$\delta(q_1, \varepsilon, z_0) = (q_1, \varepsilon)$.

5. Obtain a PDA to accept the language $L(w) = \{w / w \in (a+b)^* \; n_{(a)} w > n_{(b)} w\}$ by a final state

$= \delta(q_0, a, z_0) = (q_0, az_0)$

$\delta(q_0, b, z_0) = (q_0, bz_0)$

$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, a, b) = (q_0, \varepsilon)$

$\delta(q_0, b, b) = (q_0, bb)$

$\delta(q_0, b, a) = (q_0, \varepsilon)$

$\delta(q_0, \varepsilon, a) = (q_1, a)$

6. Obtain a PDA to accept the language $L(\omega) = \{a^n b^{2n}, n \geq 1\}$ by a final state.

=

$\delta(q_0, a, z_0) = (q_0, az_0)$

~~$\delta(q_0, a, z_0) = (q_{0,0})$~~

$\delta(q_0, a, a) = (q_0, aaa)$

$\delta(q_0, b, a) = (q_1, \varepsilon)$

$\delta(q_1, b, a) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, z_0) = (q_2, z_0)$

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1. Obtain PDA to accept the language $L(w) = \{ww^R / w \in (a+b)^*\}$ by a final state

$$\delta(q_0, a, z_0) = (q_0, az_0)$$
$$\delta(q_0, b, z_0) = (q_0, bz_0)$$

$$\delta(q_0, a, a) = (q_0, aa)$$
$$\delta(q_0, a, b) = (q_0, ab)$$

$$\delta(q_0, b, a) = (q_0, ba)$$
$$\delta(q_0, b, b) = (q_0, bb)$$

$$\delta(q_0, \varepsilon, z_0) = (q_1, z_0)$$

Assumption after half string is reached.

$$\delta(q_0, a, a) = (q_1, \varepsilon)$$
$$\delta(q_0, b, b) = (q_1, \varepsilon)$$

~~$\delta(q_1, \varepsilon, z_0) =$~~

$$\delta(q_1, a, a) = (q_1, \varepsilon)$$
$$\delta(q_1, b, b) = (q_1, \varepsilon)$$

$$\delta(q_1, \varepsilon, z_0) = (q_2, z_0)$$

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Deterministic and Non-Deterministic PDA

Let $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$ be a PDA, the PDA is deterministic if

* ~~$\delta(q_0, a, z)$~~.
* $\delta(q, a, z)$ has only one element
* if $\delta(q, \varepsilon, z)$ is not empty then $\delta(q, a, z)$ should be empty.

1.

is deterministic?